A Compilation of Problems to Unrust My Brain

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My brain has been getting a little bit rusty lately, so I decided to warm it back up.

The moment of inertia of a hollow sphere with delta functions!

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So, the moment of inertia of a hollow sphere is allegedly

$$I = \frac{2}{3}MR^2.$$

Let's derive it. We know that in general,

$$I = \int r^2 dm, \tag{1}$$

where r is the Euclidean distance to the axis of rotation and m is just the mass. Unfortunately, my brain was so rusted up that this took a hot second to rationalize. Anyway, we need two things: r, and a way to rewrite dm. Let's do the latter. We know that

$$M = \int dm = \int \rho \, dV.$$

Since we want to play with delta functions, we also observe that stuff like this is true:

$$\int \delta(r) \, dr = \int \delta^3(\mathbf{r}) \, d^3\mathbf{r} = 1. \tag{2}$$

From this we can conclude that the spatial delta function has units of inverse length. Now, we also realize that all the mass of the sphere is going to be concentrated when $r = R$. Perfect for the delta function! With all of this in mind, let's write $\rho = \sigma\delta(R - r)$. We can use two approaches to find σ, one much easier than the other. The easier way is literally just

$$\sigma = M/A = \frac{M}{4\pi R^2},$$

where A is the surface area of the sphere. The other way is to integrate:

$$M = \int \rho \, dV = \int \sigma\delta(R - r) r^2 \, dr \, d\Omega = 4\pi R^2 \sigma.$$

Ok, so now that we have ρ, let's stuff it into our inventory for use later. Our next task is to find r. If we imagine any cross section of the sphere, we just have a disk whose points are located $r\sin\theta$ from the axis of rotation. Therefore, $r \mapsto r\sin\theta$, and we thus have that

$$I = \int r^2 \sin^2\theta \sigma\delta(R - r) dV \tag{3}$$

substituting into equation 1. What's left is to simply integrate equation 3, which is done through the following:

$$\begin{aligned} I &= \int r^2 \sin^2 \theta \sigma \delta(R-r) dV \\ &= \sigma \int r^2 \sin^2 \theta \delta(R-r)\, r^2 \sin\theta dr\, d\theta\, d\phi \\ &= 2\pi\sigma \int_0^R r^4 \delta(R-r)\, dr \int_0^\pi \sin^3\theta\, d\theta. \end{aligned}$$

With the nifty property that

$$\int f(x)\delta(x-a)\, dx = f(a), \tag{4}$$

along with the fact that

$$\int_0^\pi \sin^3\theta\, d\theta = \int_0^\pi \sin\theta \left(1-\cos^2\theta\right) d\theta = 2 - \int_{-1}^1 u^2\, du = \frac{4}{3},$$

we can simplify the integrals above into

$$I = 2\pi\sigma R^4 \frac{4}{3} = \frac{8\pi}{3} R^4 \frac{M}{4\pi R^2} = \frac{2}{3} M R^2.$$

Yay!

Solving the Basel Problem with Complex Analysis

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We know the famous result of the Basel Problem

$$\sum_{n=1}^{\infty} \frac{1}{n^2} = \frac{\pi^2}{6}.$$

There are many clever ways to approach this problem, but I decided to rederive the result using complex analysis. We can rewrite this sum as an integral:

$$\sum_{n=1}^{\infty} \frac{1}{n^2} = \frac{1}{2i} \int_C \frac{1}{z^2 \tan(\pi z)} dz \tag{5}$$

On first glance, this seems anything but intuitive. However, we know the Residue Theorem, which states that for a contour γ that encapsulates a set of poles P, we have that

$$\oint_\gamma f(z) dz = 2\pi i \sum_{k \in P} \mathrm{Res}_k\, f(z) \tag{6}$$

If we make it such that all of our poles occur at integral values, one might see how we can tease the Basel problem from this relationship. The cotangent function is a good candidate for this, as $\cot(\pi z)$ has simple poles at all integers. Let's inspect the contour C in question, which is shown in the figure below.

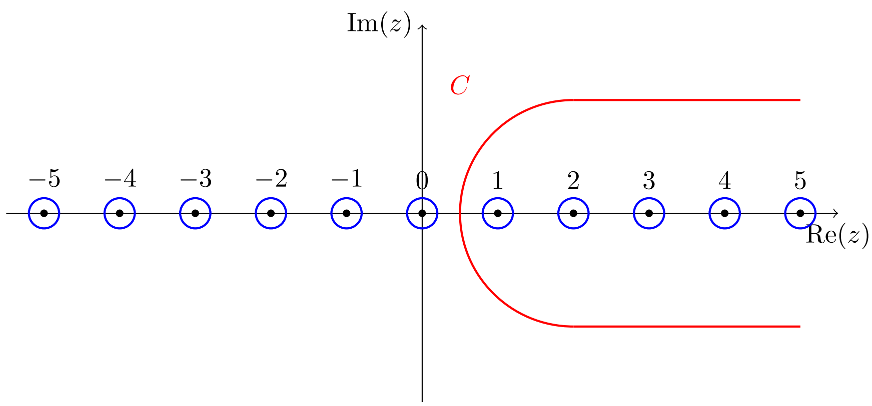

Figure 1: Our contour C that, when integrated over, will give us the solution to the Basel problem.

If we want to establish a stronger association between 5 and 6, we see that we'd like the residues to be $\mathrm{Res}_k\, f(z) = 1/\pi k^2$. We can show that this is the case. At each $z = k, k \in \mathbb{Z}$, we have a simple pole. Therefore, we can use the simple pole trick:

$$\text{If } f(z) = p(z)/q(z), \text{ then } \mathrm{Res}_k\, f(z) = p(k)/q'(k).$$

Doing some derivatives, we find that $q'(z) = z^2 \pi \sec^2(\pi z) + 2z \tan(\pi z)$. The right hand term vanishes upon evaluation because $\tan(\pi k) = 0$, while $\sec^2(\pi k) = 1$, and hence $\mathrm{Res}_k\, f(z) = 1/\pi k^2$.

Ok, so we've established that 5 is true. So how do we actually evaluate the integral? With an inspiration from Reif, we can construct another contour C' that is C reflected across the imaginary axis, then sew them together until we get something like a double keyhole contour. This is topologically equivalent to a small circle going clockwise around the pole at 0. We also know that since $f(z) \propto z^{-2}$, the extra stuff not in C and C' will vanish when we take the limit $|z| \to \infty$. Using the topological equivalence, we can shrink the contour, calling it β, and we now have that

$$\frac{1}{2i}\int_C \frac{1}{z^2\tan(\pi z)}dz = -\frac{1}{4i}\int_{|z|=\varepsilon} \frac{1}{z^2\tan(\pi z)}dz \tag{7}$$

Notice how we have acquired a negative sign and another factor of 2 in the denominator. The negative sign is there to account for the clockwise orientation of the contour, and the factor of 2 is because

$$\int_C = \frac{1}{2}\int_{C\cup C'} = \frac{1}{2}\int_\beta,$$

where the integrand is implicitly defined to be $f(z)$ as written above. We are now left with just one residue to calculate, and applying the Residue Theorem again, we have that

$$\int_{|z|=\varepsilon} \frac{1}{z^2\tan(\pi z)}dz = 2\pi i\, \mathrm{Res}_0\, f(z).$$

Observe that at $z = 0$, we have an order 3 pole, which is a bit of a pain to deal with. However, if we can still construct some $\phi(z) = z^3 f(z)$, we can find that $\mathrm{Res}_0\, f(z) = \frac{1}{2}\phi''(0)$.

So, how do we find $\phi''(0)$? We can utilize a power series approach (analytically taking derivatives becomes miserable). Let's first find the power series expansion for $1/\tan(\pi z)$, observing that it has simple poles due to $\tan'(\pi z) \neq 0$:

$$\begin{aligned}
\text{Let } \frac{1}{\tan(\pi z)} &= \frac{b_1}{z} + a_0 + a_1 z + O(z^2).\\
\text{Since } \tan(\pi z) &= \pi z + \frac{1}{3}\pi^3 z^3 + O(z^5),\\
1 &= \left(\pi z + \frac{1}{3}\pi^3 + z^3 + \dots\right)\left(\frac{b_1}{z} + a_0 + a_1 z + \dots\right)\\
&= \pi b_1 + \pi a_0 z + \left(\pi a_1 + \frac{\pi^3}{3}b_1\right)z^2 + O(z^3)\\
\implies b_1 = \frac{1}{\pi}, a_0 = 0,\ &\text{and } \pi a_1 + \frac{\pi^3}{3}b_1 = 0 \implies a_1 = -\frac{\pi}{3}.
\end{aligned}$$

If we so desired, we could calculate more terms in the expansion, but as we will see, this is sufficient. Now, writing $\phi(z) = z^3 f(z) = z/\tan(\pi z)$, we have

$$\phi(z) = b_1 + a_0 z + a_1 z^2 + \dots.$$

Therefore, $\phi''(0) = 2a_1 = -\frac{2\pi}{3}$. As such, $\mathrm{Res}_0\, f(z) = \frac{1}{2}\phi''(0) = -\pi/3$. Combining everything together, we have that

$$\sum_{n=1}^{\infty}\frac{1}{n^2} = \frac{1}{2i}\int_C \frac{1}{z^2\tan(\pi z)}dz = -\frac{1}{4i}2\pi i\,\mathrm{Res}_0\, f(z) = -\frac{\pi}{2}\cdot -\frac{\pi}{3} = \frac{\pi^2}{6},$$

which solves the Basel problem!